Dimensionality Reduction on the SPD Manifold: A Comprehensive Comparative Study Between Conventional and Geometry-Aware Methods

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Abstract. The space of Symmetric Positive Definite (SPD) matrices has gained prominence as a powerful representation in several fields such as computer vision and neuroimaging. However, the non-Euclidean geometry of the manifold formed by SPD matrices introduces a challenge, specifically in the presence of high-dimensional data, where the use of a dimensionality reduction technique is often needed. Dimensionality reduction methods typically assume the linearity of the data space, making them theoretically inappropriate for SPD matrices. To address this, researchers have increasingly adapted dimensionality reduction methods by exploiting the Riemannian structure of the SPD space. Despite their theoretical soundness, a comprehensive comparison between conventional (linear) and geometry-aware (non-linear) dimensionality reduction methods under varying data conditions is still lacking. In practice, different aspects such as the convexity of the SPD space may negatively affect the effectiveness of non-linear methods. As such, this work fills this gap through the evaluation of both linear and non-linear dimensionality reduction techniques under a binary classification setting. To that end, a dataset that presents different distributional configurations is synthetically generated. This evaluation study is also complemented by experiments performed on a real-world dataset called COBRE Connectome. The results indicate that linear and non-linear methods are comparable when applied to SPD matrices. Our analysis suggests that this may be due to two main factors: the convex nature of the SPD space and numerical issues.

Keywords: Riemannian Geometry · SPD Manifolds · Dimensionality Reduction · Functional Connectomes · Classification · Non-Euclidean Learning

1 Introduction

Symmetric Positive Definite (SPD) matrices have emerged as a central representation for structured data in domains ranging from computer vision to medical imaging and neuroscience [19,20,23,26,30]. Their ability to encode second-order statistics,

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covariance relationships, or brain connectivity patterns makes them particularly valuable in several classification tasks such as image classification [9] and action recognition [18]. However, these matrices do not belong to a linear space; they reside on a non-Euclidean Riemannian manifold. This manifold geometry introduces fundamental challenges when applying standard machine learning techniques, particularly for tasks such as dimensionality reduction. Traditional dimensionality reduction methods [12,21] assume a Euclidean structure and operate through linear projections. Although often used in practice, even after flattening SPD matrices, such approaches are not theoretically valid in the SPD context. Flattening destroys the intrinsic geometry of the manifold and may lead to non-SPD outputs, potentially distorting the statistical and physical meaning of the data [14,19,20,26].

To address these limitations, recent advances in differential geometry have led to the development of specialized dimensionality reduction techniques that preserve the Riemannian structure of SPD matrices [15,19]. However, a comprehensive comparative analysis between traditional linear methods and non-linear approaches in the context of SPD matrices remains underexplored. Hence, we have proposed in our previous work [2] a benchmark framework for comparing four dimensionality reduction strategies, including both conventional and geometry-aware ones [15,19,21,33].

Our experiments focused on synthetically generated SPD matrices with varying degrees of class separation. Interestingly, geometry-aware approaches did not always outperform simpler linear baselines such as PCA. We identified two potential causes: (i) the convexity of the SPD cone, which may reduce the observable impact of curvature in certain regimes, and (ii) numerical instability near the boundary of the manifold during logarithmic mappings.

Despite the theoretical interest of these findings, their scope was limited to controlled synthetic settings, without any validation on real-world data. In particular, no empirical evaluation was conducted on SPD matrices derived from real applications such as neuroimaging. Yet, clinical applications often involve low sample sizes, noisy measurements, and SPD matrices extracted from fMRI-derived brain connectomes. In such contexts, understanding how different dimensionality reduction methods behave is highly relevant. In this extended version, we address this gap through several key contributions:

- We incorporate an empirical evaluation of all methods on functional brain connectivity data derived from the *MLSP 2014 Schizophrenia Challenge*.
- We provide a computational complexity analysis comparing the training and inference times of all methods.
- We expand the literature review on manifold-aware dimensionality reduction.
- We benchmark two additional dimensionality reduction methods.

The remainder of this article is organized as follows. Section 2 presents an extended review of related work on geometry-aware (non-linear) dimensionality reduction for SPD matrices. Section 3 recalls the mathematical background related to the geometry of the SPD manifold. Section 4 details the dimensionality reduction techniques and the proposed evaluation protocol. Section 5 reports the results on the synthetically generated data. Section 6 presents experiments on the real-world dataset called brain connectomes. Section 7 analyzes the computational complexity of each method. Finally,

Sect. 9 concludes this work by providing a summary of findings and directions for future research.

2 Related Work

Dimensionality reduction techniques have played a pivotal role in handling high-dimensional data in numerous application fields, including image recognition, text mining, bioinformatics, and signal processing [24,32]. These techniques aim to reduce the dimensionality of feature spaces while preserving the most relevant information, thereby improving computational efficiency and mitigating the curse of dimensionality [5,11,16]. Despite being extremely useful, conventional dimensionality reduction techniques are conceptually designed for data lying in Euclidean spaces. Nevertheless, informative features might belong to non-linear spaces. For instance, covariance matrices have shown their interest in several applications such as object detection [31], texture classification [30], object tracking, action recognition and face recognition [18,20] due to their geometric and statistical properties. These matrices lie on the Riemannian manifold of SPD matrices, denoted as $\mathcal{S}_n^{++}$, which challenges the applicability of traditional Euclidean methods. In fact, linear techniques such as PCA [21] or LDA [13] flatten the SPD structure and fail to preserve the intrinsic geometry, potentially leading to non-valid outputs or distorted statistical interpretations [19,26]. In response to this issue, tangent PCA (also referred to as Principal Geodesic Analysis) was proposed [15], which linearizes the manifold locally around the Fréchet mean and applies classical PCA in the tangent space. However, when data are distributed over regions with high curvature or near the boundary of the SPD cone, this approximation can lead to distortions.

To overcome these limitations, several geometry-aware methods have emerged. Harandi et al. [19] proposed supervised dimensionality reduction based on optimization on Grassmann manifolds, using metrics such as the affine-invariant Riemannian metric (AIRM), Stein divergence, and Log-Euclidean distance. Other unsupervised non-linear methods—including Isomap [29], LLE [27], Laplacian Eigenmaps [5], and diffusion maps [11]—have also been extended to the SPD setting via Riemannian reformulations [23].

Kernel-based approaches have also gained attention by embedding the manifold into a Reproducing Kernel Hilbert Space (RKHS), allowing the use of standard machine learning tools. Notably, the use of Gaussian RBF kernels on $\mathcal{S}_n^{++}$ has enabled the application of kernel PCA, kernel SVM, and manifold-aware clustering [23]. However, ensuring positive definiteness of the kernel remains a critical constraint. Some earlier methods relied on approximations (e.g., flipping negative eigenvalues), which introduce artifacts and scale poorly with large kernel matrices.

While the literature offers a rich variety of dimensionality reduction techniques tailored to the SPD manifold, most studies evaluate these techniques in isolation, under varying experimental conditions, datasets, or performance metrics. This fragmentation makes it difficult to draw general conclusions about the comparative advantages of each approach. To our knowledge, there exists no unified benchmark that systematically compares linear and non-linear dimensionality reduction approaches under consistent

settings. This paper addresses this gap by providing a controlled and fair evaluation across representative techniques.

3 Mathematical Background: The Riemannian Geometry of SPD Matrices

The space of $n \times n$ Symmetric Positive Definite (SPD) matrices, denoted $\mathcal{S}_n^{++}$, arises in numerous fields where second-order statistics are required, including computer vision, medical imaging, and signal processing. A matrix $\mathbf{P} \in \mathbb{R}^{n \times n}$ is said to be SPD if $\mathbf{v}^\top \mathbf{P} \mathbf{v} > 0$ for all non-zero vectors $\mathbf{v} \in \mathbb{R}^n$. As a set, $\mathcal{S}_n^{++}$ forms the interior of a convex cone in the space of real symmetric matrices.

To properly operate on SPD matrices, one must consider their manifold structure when equipped with a Riemannian metric. This manifold is smooth and differentiable, with a tangent space $T_{\mathbf{P}}\mathcal{S}_n^{++}$ at each point $\mathbf{P} \in \mathcal{S}_n^{++}$. A Riemannian metric induces inner products on tangent spaces and allows the definition of intrinsic operations such as distances, geodesics, and means. Below, we briefly recall the key elements used throughout this work.

3.1 Exponential and Logarithmic Maps

Given a reference point $\mathbf{X} \in \mathcal{S}_n^{++}$, the exponential and logarithmic maps respectively transfer vectors between the manifold and the tangent space $T_{\mathbf{X}}\mathcal{S}_n^{++}$. These are respectively defined as follows,

$$\exp_{\mathbf{X}}(\mathbf{V}) = \mathbf{X}^{1/2} \exp\left(\mathbf{X}^{-1/2} \mathbf{V} \mathbf{X}^{-1/2}\right) \mathbf{X}^{1/2}, \tag{1}$$

$$\log_{\mathbf{X}}(\mathbf{P}) = \mathbf{X}^{1/2} \log\left(\mathbf{X}^{-1/2} \mathbf{P} \mathbf{X}^{-1/2}\right) \mathbf{X}^{1/2}, \tag{2}$$

where $\mathbf{P} \in \mathcal{S}_n^{++}$, $\mathbf{V} \in T_{\mathbf{X}}\mathcal{S}_n^{++}$, and $\log(\cdot)$, $\exp(\cdot)$ denote the matrix logarithm and exponential, respectively.

3.2 Riemannian Metrics on SPD Manifolds

Several metrics have been proposed to measure distances between SPD matrices while taking into account the geometric structure of the SPD space. The following are the most commonly used ones:

– **Affine-Invariant Riemannian Metric (AIRM) [26]:**

$$d_{\text{AIRM}}(\mathbf{P}, \mathbf{Q}) = \left\| \log\left(\mathbf{P}^{-1/2} \mathbf{Q} \mathbf{P}^{-1/2}\right) \right\|_F, \tag{3}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. AIRM is invariant under congruent transformations and is well-suited for affine-invariant applications.

– **Stein Divergence** [10]:

$$d_{\text{Stein}}(\mathbf{P}, \mathbf{Q}) = \log\det\left(\frac{\mathbf{P}+\mathbf{Q}}{2}\right) - \frac{1}{2}\log\det(\mathbf{PQ}). \tag{4}$$

Stein divergence is symmetric, positive-definite, and computationally efficient, making it attractive for large-scale problems.

– **Jeffrey Divergence:**

$$d_{\text{Jeffrey}}(\mathbf{P}, \mathbf{Q}) = \frac{1}{2}\left(\text{tr}(\mathbf{P}^{-1}\mathbf{Q}) + \text{tr}(\mathbf{Q}^{-1}\mathbf{P})\right) - n, \tag{5}$$

where $\text{tr}(\cdot)$ denotes the trace operator and n the matrix dimension.

– **Log-Euclidean Metric (LEM)** [3]**:**

$$d_{\text{LEM}}(\mathbf{P}, \mathbf{Q}) = \|\log(\mathbf{P}) - \log(\mathbf{Q})\|_F. \tag{6}$$

LEM treats SPD matrices in the logarithmic domain, allowing efficient computation while partially preserving geometric structure.

3.3 Fréchet Mean

The Fréchet mean [17], generalizes the Euclidean average to Riemannian manifolds. Given a set of SPD matrices $\{\mathbf{P}_1, \ldots, \mathbf{P}_k\}$, and a metric $d(\cdot, \cdot)$, the Fréchet mean $\mathbf{P}_\mu \in \mathcal{S}_n^{++}$ minimizes the sum of squared distances:

$$\mathbf{P}_\mu = \arg\min_{\mathbf{P}\in\mathcal{S}_n^{++}} \sum_{i=1}^{k} d^2(\mathbf{P}, \mathbf{P}_i). \tag{7}$$

When using AIRM, the Fréchet mean must be computed iteratively, typically via gradient descent on the manifold. It plays a central role in manifold learning algorithms, including Tangent PCA and clustering on SPD spaces.

4 Methods and Evaluation Protocol

This section details the dimensionality reduction techniques investigated in this work, alongside the synthetic data generation protocol designed to evaluate their behavior under controlled scenarios. We begin by describing each method—classical and geometry-aware—then proceed to the construction of synthetic SPD datasets with varying distribution overlap.

4.1 Dimensionality Reduction Techniques

We benchmark six representative techniques: three *conventional* baselines, namely, Principal Component Analysis (PCA), an Autoencoder (AE) and Kernel PCA and three *geometry-aware* methods taking into account the non-linearity of the SPD space, namely, tangent PCA (tPCA), SPD-tailored kernel PCA, and the supervised manifold-projection approach of Harandi *et al.* [19].

Conventional Methods

Principal Component Analysis (PCA). The classical PCA procedure involves the following steps:

- **Step 1 (Centering the Data):** Let $\{\mathbf{X}_i\}_{i=1}^N$ denote the dataset of vectorized SPD matrices. The mean of the data is computed as

$$\mu = \frac{1}{N}\sum_{i=1}^{N}\mathbf{X}_i, \tag{8}$$

  and each data point is centered:

$$\mathbf{X}_i' = \mathbf{X}_i - \mu. \tag{9}$$

- **Step 2 (Covariance Matrix Estimation):** The empirical covariance matrix of the centered data is computed by:

$$\mathbf{C} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{X}_i'(\mathbf{X}_i')^\top. \tag{10}$$

- **Step 3 (Eigen Decomposition):** The covariance matrix is then decomposed as:

$$\mathbf{C} = \mathbf{U}\mathbf{S}\mathbf{U}^\top, \tag{11}$$

  where $\mathbf{U}$ contains the eigenvectors (principal directions), and $\mathbf{S}$ is the diagonal matrix of eigenvalues.
- **Step 4 (Projection onto Principal Components):** Finally, each centered sample is projected onto the leading k eigenvectors:

$$\mathbf{Y}_i = \mathbf{V}^\top\mathbf{X}_i', \tag{12}$$

  where $\mathbf{V} \in \mathbb{R}^{d\times k}$ contains the top k eigenvectors.

In our experiments on synthetically generated data, PCA is directly applied by flattening each SPD matrix into a full vector of dimension $d = n^2$. In contrast, for real-world data (brain connectomes), we leverage the symmetry of SPD matrices and extract only the upper triangular part to avoid redundancy, resulting in a reduced vector of dimension $d = \frac{n(n+1)}{2}$. This choice is aligned with standard experimental protocols [6] in neuroimaging to ensure compactness and computational efficiency.

Autoencoder. Autoencoders are neural architectures designed to learn compact representations of high-dimensional data by minimizing reconstruction error. They consist of two main components: an encoder that maps inputs to a low-dimensional latent space, and a decoder that reconstructs the original data from this latent representation. The model is trained in an unsupervised manner to minimize the discrepancy between input and output, typically using the Mean Squared Error (MSE) loss. In our setting, each SPD matrix is flattened into a vector of dimension $d = n^2$, discarding its symmetry structure. This transformation enables the use of standard feedforward layers but

neglects the non-Euclidean geometry of SPD manifolds. The encoder is composed of four fully connected layers with ReLU activations, progressively reducing the dimensionality from d to 2. The decoder mirrors this structure to reconstruct the original input. No activation is applied at the output layer to preserve the numerical range. The autoencoder is trained using the Adam optimizer for 100 epochs with a batch size of 32. We apply the autoencoder to flattened SPD matrices of size $n \times n$, resulting in an input dimension of $d = n^2$. Only 2D embeddings are extracted and evaluated for classification. For real-world data from the MLSP 2014 Schizophrenia Challenge, each functional connectivity matrix of size 28×28 is flattened into a vector of length 784. The same architecture is used to obtain 2D embeddings for downstream evaluation.

Kernel PCA. is a classical dimensionality reduction method that extends PCA by applying the kernel trick to project data into a high-dimensional feature space before applying PCA. In our experiments, we apply this method directly on flattened SPD matrices, without accounting for the Riemannian structure of the manifold. We use the Gaussian (RBF) kernel as in the original formulation proposed by Schölkopf et al. [28].

Geometry-Aware Techniques

Tangent PCA (tPCA). Tangent PCA [14] extends PCA to Riemannian manifolds by linearizing the data at a reference point. The procedure involves:

1. Computing the Fréchet mean $\mu \in \mathcal{S}_n^{++}$ of the dataset under the chosen Riemannian metric (AIRM).
2. Mapping each SPD matrix $\mathbf{X}_i$ to the tangent space $T_\mu \mathcal{S}_n^{++}$ via the logarithmic map $\log_\mu(\mathbf{X}_i)$.
3. Performing standard PCA in the Euclidean tangent space.
4. Projecting back to the manifold using the exponential map.

This method preserves local manifold geometry and is less prone to the distortion introduced by flattening.

Harandi's Manifold-Based Subspace Learning. The method proposed by Harandi et al. [19] learns a projection matrix $\mathbf{W} \in \mathbb{R}^{n \times m}$, where $m < n$, such that each SPD matrix $\mathbf{X} \in \mathcal{S}_n^{++}$ is mapped to $\mathbf{W}^\top \mathbf{X} \mathbf{W} \in \mathcal{S}_m^{++}$. The objective is to preserve intra-class similarity while maximizing inter-class separability using graph-based discriminant analysis.

The optimization problem is defined over the Grassmannian manifold $\mathcal{G}(m, n)$, subject to the orthonormality constraint $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_m$. The cost function incorporates affinity graphs constructed from pairwise similarities under several metrics, including, Affine-Invariant Riemannian Metric (AIRM), Stein divergence, Jeffrey divergence, Log-Euclidean metric, and the Euclidean metric [4, 10, 26].

The optimization is performed using conjugate gradient descent on the manifold. This approach is computationally intensive but benefits from stronger geometric guarantees.

SPD-Tailored Kernel PCA. SPD-tailored Kernel Principal Component Analysis is an extension of kernel PCA that is extended to the SPD manifold [28]. Let $\{X_i\}_{i=1}^N \subset \mathcal{S}_n^{++}$, where $\mathcal{S}_n^{++}$ denotes the manifold of $n \times n$ symmetric positive-definite (SPD) matrices. Rather than flattening each X_i or projecting it onto a tangent space, we embed the manifold into an RKHS via a *geometry-aware* kernel

$$k : \mathcal{S}_n^{++} \times \mathcal{S}_n^{++} \longrightarrow \mathbb{R},$$

which: (i) is strictly positive definite and (ii) depends only on a chosen Riemannian distance d between SPD matrices:

$$k_\sigma(X_i, X_j) = \exp\left(-\frac{d^2(X_i, X_j)}{2\sigma^2}\right), \qquad \sigma > 0. \tag{13}$$

Several choices of d are valid. We adopt the Log-Euclidean metric [4]

$$d_{\mathrm{LE}}(X_i, X_j) = \|\log X_i - \log X_j\|_{\mathrm{F}}, \tag{14}$$

which yields a provably positive-definite Gaussian kernel on $\mathcal{S}_n^{++}$ [22,23].

Dual Formulation. Define the Gram matrix $K \in \mathbb{R}^{N\times N}$ with entries $K_{ij} = k_\sigma(X_i, X_j)$. After centering K in feature space, K–PCA solves the eigenproblem $Kc_\ell = \lambda_\ell c_\ell$, $\ell = 1, \ldots, L$, where $\{\lambda_\ell\}$ are the largest eigenvalues. The L-dimensional embedding of X_i is then

$$z_i = \left(\sqrt{\lambda_1}\, c_{1,i}, \ldots, \sqrt{\lambda_L}\, c_{L,i}\right)^\top \in \mathbb{R}^L. \tag{15}$$

Because the kernel in Eq. (13) respects the manifold geometry, the coordinates defined in Eq. (15) constitute a *linear* Euclidean representation that preserves non-linear relationships on $\mathcal{S}_n^{++}$ without resorting to local tangent approximations [23]. Therefore, these embeddings can be provided directly to standard classifiers such as k-NN in $\mathbb{R}^L$.

4.2 Synthetic SPD Data Generation

To benchmark the methods under controlled conditions, we design three synthetic datasets with varying degrees of class separability. Each class is modeled as a distribution of SPD matrices generated from samples in the tangent space.

Sampling Protocol. Each matrix $\mathbf{A}_i$ is generated as:

$$\mathbf{A}_i = \exp(\mathbf{S}_i), \tag{16}$$

where

$$\mathbf{S}_i = \frac{1}{2}(\mathbf{M}_i + \mathbf{M}_i^\top), \quad \mathbf{M}_{i,j} \sim \mathcal{U}(a, b), \tag{17}$$

with $\mathcal{U}(a, b)$ denoting the uniform distribution over the interval $[a, b]$. The exponential map ensures that $\mathbf{A}_i \in \mathcal{S}_n^{++}$. Each class is generated by choosing different ranges $[a, b]$ to control inter-class distance.

Distribution Configurations. We define three configurations:

– **Case 1 (Distant Distributions):**

$$\text{Class 1: } a = 1, b = 2; \quad \text{Class 2: } a = 0, b = 1$$

– **Case 2 (Moderate Overlap):**

$$\text{Class 1: } a = 0, b = 1; \quad \text{Class 2: } a = 0.25, b = 1.25$$

– **Case 3 (Strong Overlap):**

$$\text{Class 1: } a = 0, b = 1; \quad \text{Class 2: } a = 0.125, b = 1.125$$

In each case, we generate 1000 SPD matrices per class, resulting in a balanced dataset of 2000 samples. These scenarios allow us to assess how methods perform as class separation decreases.

Visualization of Synthetic Distributions. To illustrate the geometry of the generated datasets, we project the matrices from each configuration onto two-dimensional space. Figure 1 shows the spatial layout of class distributions. As expected, Case 1 yields well-separated clusters, while Case 3 exhibits heavy inter-class entanglement. This visualization provides qualitative insight into the classification difficulty and geometric structure in each scenario.

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5 Experiments and Results

5.1 Experimental Setup

We evaluate the performance of dimensionality reduction techniques in classifying SPD matrices under three configurations of data overlap (distant, slightly interwoven, and strongly interwoven). For each method, we apply a 2D embedding followed by k-Nearest Neighbors (k-NN) classification. We report the mean and standard deviation of accuracy over 10 random splits (80% training/20% testing). Dimensionality reduction is performed using PCA, tangent PCA, Kernel PCA, SPD-tailored Kernel PCA, autoencoder, and Harandi's method. We leverage the *Geomstats* library for the generation and manipulation of SPD matrices, which provides a robust framework for geometric learning and processing on Riemannian manifolds [25]. Additionally, for testing Harandi's method, we use the official code released by the authors [7, 19].

5.2 Results on Synthetic Data

Table 1 presents the classification accuracies across the three configurations. The evaluation of PCA, Tangent PCA, Autoencoder, and Harandi's method follows the same experimental protocol described in the previous study [2]. In this extended version,

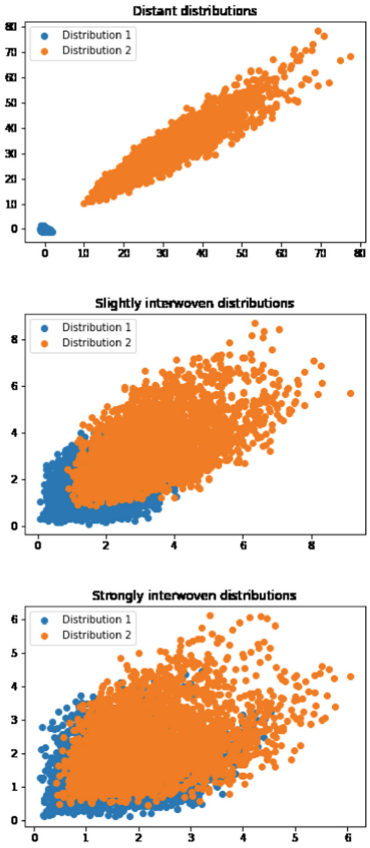


Fig. 1. Visualization of synthetic SPD distributions. From top to bottom: (1) well-separated, (2) moderately overlapping, and (3) highly overlapping distributions [2].

two additional techniques are introduced: standard Kernel PCA and SPD-tailored Kernel PCA. In the well-separated case (Case 1), all methods achieve perfect separation (100%). However, in Case 2 (slightly interwoven), the autoencoder slightly outperforms other techniques with a mean accuracy of 89.42%, followed closely by Tangent PCA (88.20%). Harandi's method reaches 86.69%, and PCA 88.03%. In Case 3 (strongly interwoven), the performance drops for all methods, but the Autoencoder remains the most robust (70.80%), followed by PCA and Tangent PCA (around 69%), while Harandi's approach underperforms at 65.67%. Furthermore, SPD-tailored Kernel PCA performs competitively across all cases, matching or surpassing the performance of the autoencoder in Case 2 (89.75%) and maintaining stability in more challenging scenarios. In contrast, classical Kernel PCA underperforms, especially in the strongly interwoven case (64.55%), highlighting its limitations when applied directly to flattened SPD data without accounting for manifold geometry.

Table 1. Comparison of Method Performance Across Different Cases.

Case	Method	Mean Accuracy (%)	Standard Deviation (σ)
Case 1: Distant Distributions	Tangent PCA	100.00	-
	Classic PCA	100.00	-
	Autoencoder	100.00	-
	Harandi's Method	100.00	-
	SPD-tailored Kernel PCA	100.00	-
	Kernel PCA	98.97	0.68
Case 2: Slightly Interwoven	Tangent PCA	88.20	1.09
	Classic PCA	88.03	1.56
	Autoencoder	89.42	1.67
	Harandi's Method	86.69	1.86
	SPD-tailored Kernel PCA	89.75	1.00
	Kernel PCA	78.53	3.09
Case 3: Strongly Interwoven	Tangent PCA	69.33	1.71
	Classic PCA	69.40	1.99
	Autoencoder	70.80	1.13
	Harandi's Method	65.67	3.10
	SPD-tailored Kernel PCA	68.62	3.12
	Kernel PCA	64.55	2.53

5.3 Influence of the Riemannian Metric (Harandi)

Table 2 reports the performance of Harandi's method under various metrics. The Log-Euclidean metric offers the most stable performance across all configurations. In highly overlapping distributions (Case 3), it even outperforms AIRM, which is traditionally favored for its affine invariance. Stein and Jeffreys also perform comparably well, while the Euclidean variant shows clear degradation under complex data configurations. These findings reinforce the importance of metric selection in Riemannian learning settings.

Table 2. Performance of Harandi's Method Across Different Metrics [2].

Case	AIRM		Stein		Jeffreys		Log-Euclidean		Euclidean	
	Mean (%)	Std. Dev (σ)	Mean (%)	Std. Dev (σ)	Mean (%)	Std. Dev (σ)	Mean (%)	Std. Dev (σ)	Mean (%)	Std. Dev (σ)
Case 1	100.00	-	100.00	-	100.00	-	100.00	-	100.00	-
Case 2	86.56	1.47	**86.69**	1.86	86.66	1.48	86.03	1.60	85.74	1.61
Case 3	65.10	3.42	65.47	2.95	65.49	2.24	**65.67**	3.10	62.49	3.58

Figure 2 shows the 2D visualization of the SPD matrices after applying the various dimensionality reduction techniques, demonstrating that the distribution configurations are preserved, highlighting the need for approaches that can better handle overlapping distributions.

5.4 Classification Error According to the Eigenvalue Range

To understand how the geometry of the SPD manifold affects classification performance, we analyze the classification error with respect to the minimum eigenvalue of each SPD matrix. This analysis aims to explore the relationship between the geometric structure of the SPD cone and the effectiveness of dimensionality reduction techniques, in particular classic PCA and Tangent PCA.

The minimum eigenvalue serves as an indicator of how close a matrix lies to the boundary of the SPD cone. Matrices with smaller minimum eigenvalues are closer to the boundary, where non-linear curvature is more pronounced, whereas matrices with larger minimum eigenvalues reside in regions that are more locally Euclidean due to the convex nature of the manifold.

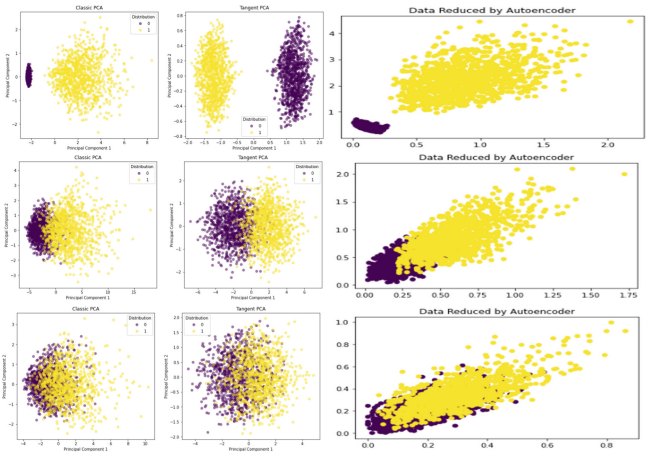


Fig. 2. 2D Visualization of SPD matrices after dimensionality Reduction [2].

As shown in Fig. 3, we partition the dataset into bins according to the minimum eigenvalue and report the classification errors within each bin for Case 2 and Case 3. The results reveal that classification error is generally lower in the bins associated with higher minimal eigenvalues, suggesting that classic PCA performs better in locally linear regions of the SPD space. Interestingly, classic PCA also slightly outperforms Tangent PCA in the bins corresponding to very low eigenvalues, which could be attributed to numerical instability when computing the matrix logarithm near the cone boundary. Conversely, Tangent PCA demonstrates superior performance for moderately low eigenvalues.

In summary, while classic PCA excels in certain scenarios, Tangent PCA's performance is notably influenced by the geometric properties of the data. This study highlights the importance of understanding the underlying structure of data distributions when selecting dimensionality reduction techniques, particularly in complex scenarios where traditional methods may struggle.

6 Real-World Application: Brain Connectomes

To validate that our findings on synthetic data translate to clinically relevant scenarios, we replicate the entire evaluation pipeline on a public neuroimaging cohort. In what follows, we describe the dataset and the experimental results.

6.1 Dataset Description

The *MLSP 2014 Schizophrenia Classification Challenge* [8] is a public dataset collected at the *Mind Research Network* as part of the NIH-funded Center of Biomedical Research Excellence (COBRE) initiative.

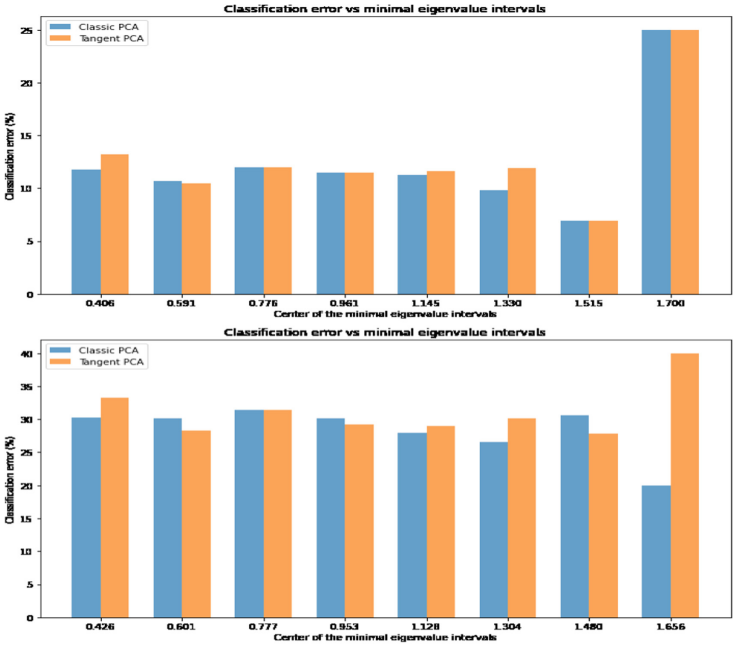


Fig. 3. Classification errors as a function of minimum eigenvalue ranges [2].

The dataset comprises resting-state functional magnetic resonance imaging (rs-fMRI) scans from **86 subjects**, including both healthy controls and patients diagnosed with schizophrenia. From these scans, *Functional Connectivity Networks* (FCNs) were computed based on the temporal correlations between the activation time series of **28 predefined brain Regions of Interest (ROIs)**. Each FCN is represented as a 28×28 **symmetric positive-definite (SPD) matrix**, summarizing the pairwise synchrony between the selected ROIs over time. As such, these matrices lie on the Riemannian manifold $\mathcal{S}_{++}^{28}$ of SPD matrices, a geometric structure that naturally accommodates the non-Euclidean properties of correlation-based functional connectivity.

In line with our study objective, we focus solely on the **Functional Network Connectivity (FNC)** features, a functional modality capturing global synchrony between independent brain regions. These features quantify the synchrony between spatially independent brain networks, computed via correlations between timecourses extracted through group Independent Component Analysis (GICA) on rs-fMRI data. Each FCN thus provides a subject-level summary of functional organization in the brain, and is widely used as a biomarker in schizophrenia research [1].

Figure 4 shows a visual comparison of the FCNs corresponding to a healthy control and a schizophrenic patient. As FCNs are symmetric and positive-definite by construction, they reside in a non-Euclidean geometric space, motivating the use of Riemannian tools for dimensionality reduction and classification.

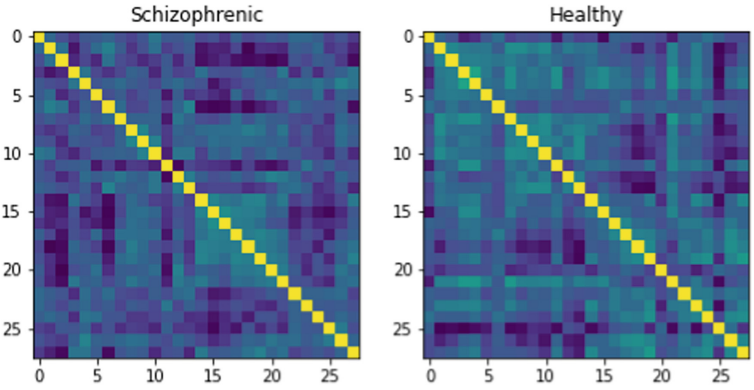


Fig. 4. Visualization of the first two brain connectomes from the MSLP dataset. The left matrix corresponds to a healthy control, while the right matrix corresponds to a schizophrenic patient. Each connectome is represented as a 28×28 correlation matrix on the SPD manifold.

6.2 Evaluation Settings

As in our experimental with synthetic data, the original SPD matrices are reduced to 2D. In addition, the reduced embeddings are evaluated through a classification step. Specifically, a k-Nearest Neighbors ($k = 5$) classifier is used on top of the reduced embeddings. For each method, we split the data into 80% for training and 20% for testing sets. The evaluation is repeated 5 times to account for variance due to sampling.

6.3 Results on Connectome Data

Table 3 reports the macro-averaged accuracy for all methods. A finer ablation of Harandi's method across five Riemannian metrics is given in Table 4.

Table 3. Classification performance of dimensionality reduction methods.

Method	Accuracy (%)
PCA	**61.11**
Tangent PCA	55.56
Harandi (AIRM)	55.89
Autoencoder	50.00
SPD-tailored kernel PCA	50.00
Kernel PCA	50.00

In this real-world classification task, linear methods demonstrate comparable performance to their geometry-aware counterparts. PCA achieves the highest accuracy at **61.11%**, tangent PCA follows closely with an accuracy of **55.56%**, Harandi's method performs similarly (around **55.89%**), regardless of the chosen metric, yet suffers from high variability across trials (standard deviations up to 15.6%). Moreover, the autoencoder and both variants of kernel PCA underperform with accuracy levels of **50%**. These results are consistent with the findings observed in our experiments on synthetic data, where geometry-aware methods do not consistently outperform classical Euclidean approaches.

Table 4. Classification accuracy (%) of Harandi's method across different Riemannian metrics in 2D embeddings. Results are reported as mean $\pm$ standard deviation over 10 runs.

Metric	Accuracy (mean $\pm$ std)
AIRM	55.89 ± 9.20
Stein	54.12 ± 9.77
Jeffrey	54.72 ± 14.18
Log-Euclidean	43.53 ± 15.61
Euclidean	55.29 ± 13.70

7 Computational Analysis: Empirical Training and Inference Time

Dimensionality reduction techniques differ not only in classification accuracy but also in computational efficiency. This section compares methods across the empirical runtime.

Table 5. Training and inference time (in seconds) for all methods on synthetic and real data.

Method	Synthetic Data (3×3)		Real Data (28×28)	
	Train Time (s)	Inference Time (s)	Train Time (s)	Inference Time (s)
PCA	0.0034	0.0000	0.0310	0.0020
Tangent PCA	0.0259	0.0020	0.5028	0.0030
Autoencoder	11.4176	0.5281	12.6671	0.0041
Harandi (AIRM)	**259.6322**	**59.8862**	**9.1101**	**0.1711**
Harandi (Stein)	9.7903	2.0544	0.8974	0.0151
Harandi (Jeffreys)	12.7304	2.5598	0.9852	0.0521
Harandi (Log-Euclidean)	1.1734	0.2275	0.8727	0.0215
Harandi (Euclidean)	0.7899	0.0368	0.2335	0.0110
SPD-tailored kernel PCA	**19 413.185**	0.065	31.548	0.039
Kernel PCA	0.4707	0.3673	0.0229	0.0368

As reported in Table 5, linear methods such as PCA are the most computationally efficient, both in training and inference. Autoencoders require significantly longer training times, attributed to iterative optimization over multiple layers, while inference remains fast. Geometry-aware approaches like Harandi's method introduce additional costs due to operations on the manifold. Notably, SPD-tailored Kernel PCA incurs the highest training cost, dominated by the construction of the Log-Euclidean kernel matrix, which scales quadratically with the number of samples. These observations highlight the superiority of standard methods in terms of computational efficiency.

8 Discussion

This study provides a comprehensive comparison of classical and geometry-aware dimensionality reduction techniques for SPD-valued data, considering both synthetically generated data and real brain connectome data. Empirically, classical PCA consistently yields higher performance, even outperforming more sophisticated manifold-based approaches in low-dimensional settings. This outcome can be attributed to several factors: the convexity of the SPD cone, which may reduce the need for non-linear methods and numerical instability in logarithmic mappings near the boundary of $\mathcal{S}_n^{++}$. From a computational aspect, PCA remains the most efficient method. While tangent PCA is slightly more costly in terms of computation, Harandi's method incurs a substantial due to the calculation of pairwise distance and optimization over the Grassmannian. SPD-tailored kernel PCA exhibits an asymmetric runtime profile, rendering it impractical for large datasets. On synthetic batches, kernel construction required several hours; however, the cost dropped significantly on the smaller real-world dataset.

This study has several limitations. First, the COBRE dataset includes only 86 subjects, limiting statistical power and potentially hiding the benefits of more complex methods. Second, we focused exclusively on 2D dimensionality reduction, which may

over-compress relevant geometry and penalize methods that benefit from richer representations. Third, only functional connectivity (FNC) was considered; integrating structural modalities such as SBM or DTI could yield complementary insights.

9 Conclusion

This study presented a comprehensive comparative analysis of classical and geometry-aware dimensionality reduction techniques applied to symmetric positive definite (SPD) matrices, evaluated under both controlled synthetic scenarios and real-world neuroimaging data. While prior work has largely focused on synthetic benchmarks, our study bridges this gap by validating performance on real connectome-derived SPD data, thereby providing empirical insights into their practical applicability.

Our results demonstrate that geometry-aware methods, despite their theoretical soundness, do not systematically outperform Euclidean baselines under strong compression regimes.

Future work should investigate higher-dimensional reduction, integrate structural and functional neuroimaging modalities, and explore hybrid architectures such as SPDNet or U-SPDNet that combine deep learning with Riemannian priors. Moreover, a deeper theoretical analysis of the local geometry of the SPD manifold may help explain the empirical behaviors observed and clarify the cases in which geometry-aware approaches offer tangible advantages over classical baselines.

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